OPTICAL MUSIC RECOGNITION FOR JAZZ LEAD SHEETS: A CRNN-CTC APPROACH FOR MONOPHONIC LINES

POL CASANOVAS PUIG

Thesis supervisor
MANEL FRIGOLA BOURLON (Department of Automatic Control)

GEP tutor
XAVIER MORALES SOROLLA

Degree
Bachelor's Degree in Informatics Engineering (Computing)

GEP Deliverable 2: Temporal Planning

Facultat d'Informàtica de Barcelona (FIB)

Universitat Politècnica de Catalunya (UPC) – BarcelonaTech

Contents

1 Project Overview and Global Data

This document presents the temporal planning for the Bachelor's Thesis project *Optical Music Recognition for Jazz Lead Sheets: A CRNN-CTC Approach for Monophonic Lines.* The planning follows an agile, iterative methodology as described in GEP Deliverable 1, with development organised into five sprints plus a concurrent project management work package.

1.1 Key Dates and Effort

Table 1 summarises the global planning parameters.

Table 1: Global project data.

Parameter	**Value**
Project start date	February, 2026
Project end date	June, 2026
Total duration	21 weeks
Average daily effort	4–5 hours
Total estimated effort	**509 hours**
Number of team members	1 (student) + 1 (supervisor, advisory)

The project is carried out by a single developer (the student), with periodic advisory input from the thesis supervisor. Being a solo project, all roles researcher, developer, tester, and technical writer are fulfilled by the same person. The supervisor contributes approximately 1–2 hours per meeting, roughly biweekly.

1.2 Methodology Recap

As established in Deliverable 1, the project follows an **iterative and incremental methodology** inspired by agile principles. Rather than a rigid waterfall sequence, the core technical work (dataset construction, model development, and evaluation) is organised into **sprints** of approximately four weeks each. Each sprint produces a working, evaluated artefact and concludes with a review that informs the next iteration.

Specifically, Sprints 3 and 4 form an explicit **refinement loop**: after the initial model is trained and evaluated (Sprint 3), error analysis feeds back into data pipeline improvements and model adjustments (Sprint 4). This cycle can repeat within Sprint 4 as needed before moving to final evaluation.

A concurrent project management work package (WP0) runs throughout the entire project, encompassing documentation, supervisor meetings, and GEP-related deliverables.

2 Task Description

This section describes each task at individual task level, grouped by work package. Each task is identified by a code (e.g., **T1.3**), assigned an estimated effort in hours, and linked to its dependencies.

Resources are common to all tasks unless otherwise noted:

- **Human:** Student (Pol Casanovas Puig). Supervisor (Manel Frigola Bourlon) participates in review meetings (T0.5).
- **Hardware:** Personal computer used for all coding, training, and documentation.

- **Software:** Python 3.14, PyTorch, OpenCV, music21, LilyPond, LaTeX, Git/GitHub, among others.

2.1 WP0: Project Management

This work package runs **concurrently** with all sprints, from project start to end. It encompasses all management, documentation, and coordination activities.

T0.1 - Context and scope definition (20 h)
Research and write the project context, state-of-the-art review, alternative justification, scope definition, requirements, risks, and methodology. Corresponds to GEP Deliverable 1 content.
Dependencies: None (project start).
Resources: Standard + Zotero reference manager.

T0.2 - Temporal planning (12 h)
Define the work breakdown structure, estimate task durations, establish dependencies, and produce the Gantt chart. Corresponds to this document (GEP Deliverable 2).
Dependencies: T0.1.

T0.3 - Budget and sustainability report (15 h)
Prepare the economic budget (human resources, hardware amortisation, software licences, indirect costs) and the sustainability analysis (environmental, economic, and social dimensions). Corresponds to GEP Deliverable 3.
Dependencies: T0.2.

T0.4 - Final GEP document integration (8 h)
Compile, revise, and integrate GEP deliverables 1–3 into a coherent final management document.
Dependencies: T0.3.

T0.5 - Supervisor meetings and follow-up (25 h)
Biweekly meetings with the thesis supervisor (∼1.5 h each, including preparation and minutes), plus ad-hoc consultations. Approximately 15 meetings over 21 weeks. This task is **intermittent** throughout the entire project.
Dependencies: Concurrent with all tasks.

T0.6 - Thesis writing and documentation (50 h)
Continuous writing of the final thesis document (`main.tex`), including the introduction, methodology, experimental chapters, and conclusions. This task is **concurrent** with all sprints, intensifying during Sprint 5.
Dependencies: Concurrent with all tasks; intensifies after T4.5.

WP0 subtotal: 130 h

2.2 WP1 - Sprint 1: Literature Review and Baseline

Duration: Weeks 1–4 (February – March 2026).
Sprint goal: Establish the theoretical foundation and implement a classical computer vision baseline to quantify the difficulty of the task.

T1.1 - OMR literature survey (25 h)
Systematic review of OMR approaches: traditional pipelines [1], CRNN-CTC methods [2], and transformer-based systems [3], [4]. Document findings and annotated bibliography.
Dependencies: None (sprint start).
Resources: Standard + access to IEEE, ACM, and arXiv repositories.

T1.2 - Dataset exploration and analysis (15 h)
Download, inspect, and document publicly available OMR datasets (PrIMuS, CameraPrimus, MUSCIMA++, DeepScoresV2, OLiMPiC, GrandStaff). Analyse encoding formats, annotation quality, and suitability for monophonic training.
Dependencies: T1.1 (partial; may overlap).

T1.3 - Classical CV baseline implementation (25 h)
Implement a morphological staff-line detection and removal pipeline using OpenCV. Evaluate staff-line recall on clean typeset images (PrIMuS), distorted camera images (CameraPrimus), and handwritten Real Book scans.
Dependencies: T1.2.

T1.4 - Baseline evaluation and report (10 h)
Quantitative evaluation of the morphological baseline: staff-line recall, precision, noise artefact rate. Document findings and failure modes that motivate the deep learning approach.
Dependencies: T1.3.

T1.5 - Sprint 1 review (3 h)
Present results to supervisor, discuss lessons learned, and confirm Sprint 2 scope.
Dependencies: T1.4.

WP1 subtotal: 78 h

2.3 WP2 - Sprint 2: Dataset Construction

Duration: Weeks 5–8 (March 2026).
Sprint goal: Build a synthetic monophonic dataset in Real Book style with paired image–annotation data suitable for CRNN-CTC training.

T2.1 - Output encoding design (12 h)
Define the LMX-based symbolic vocabulary: token set (pitch + octave + duration + rests + accidentals + barlines), special tokens (clef, time/key signatures), and the CTC blank token. Write the conversion script from PrIMuS `.semantic` format to LMX.
Dependencies: T1.5.
Resources: Standard + music21 library.

T2.2 - Synthetic rendering pipeline (35 h)
Develop a LilyPond-based pipeline to render PrIMuS-derived monophonic scores in Real Book jazz style using the LilyJAZZ font. Generate ∼43,000 staff-line images (PNG) with paired LMX annotations. Handle edge cases: multi-rest tokens, clef filtering, degenerate sequences.
Dependencies: T2.1.
Resources: Standard + LilyPond with LilyJAZZ package.

T2.3 - Data augmentation pipeline (15 h)
Implement augmentation transforms (Gaussian noise, Gaussian blur, perspective warp) to produce a "scanned" variant of the dataset that simulates real-world degradation. Calibrate distortion parameters against actual Real Book scan characteristics.
Dependencies: T2.2.
Resources: Standard + OpenCV.

T2.4 - Dataset validation and split preparation (8 h)
Validate image–label alignment, filter corrupted or degenerate samples, and prepare reproducible train/validation/test splits (80/10/10). Produce dataset statistics summary.
Dependencies: T2.3.

T2.5 - Sprint 2 review (3 h)
Present the dataset to the supervisor, review quality metrics, and confirm readiness for model training.
Dependencies: T2.4.

WP2 subtotal: 73 h

2.4 WP3 - Sprint 3: CRNN-CTC Model Development

Duration: Weeks 9–12 (April 2026).
Sprint goal: Design, implement, and perform an initial training run of the CRNN-CTC model, establishing the evaluation framework.

T3.1 - CRNN architecture design and implementation (25 h)
Design the CNN backbone (convolutional feature extractor), the bidirectional LSTM sequence module, and the fully connected CTC head. Implement in PyTorch with configurable hyperparameters, keeping the model lightweight enough for training on a consumer GPU.
Dependencies: T2.5.
Resources: Standard + consumer GPU.

T3.2 - Training pipeline implementation (20 h)
Implement the training loop with CTC loss, a learning rate scheduler, gradient clipping, early stopping, and best-model checkpointing. Implement the dataset loader with dynamic padding and collation.
Dependencies: T3.1.

T3.3 - Initial training run (15 h)
Execute the first full training run. Monitor loss curves, diagnose convergence issues, and document initial baseline metrics.
Dependencies: T3.2.

T3.4 - Evaluation framework (12 h)
Implement greedy CTC decoding, Symbol Error Rate (SER) computation, per-sample error breakdown (substitutions, insertions, deletions), and worst-prediction visualisation. This task runs in partial overlap with T3.3.
Dependencies: T3.2 (may overlap with T3.3).

T3.5 - Sprint 3 review (3 h)
Analyse initial training results with the supervisor, identify dominant error modes, and plan the Sprint 4 refinement strategy.
Dependencies: T3.3, T3.4.

WP3 subtotal: 75 h

2.5 WP4 - Sprint 4: Iterative Refinement

Duration: Weeks 13–16 (April–May 2026).
Sprint goal: Close the train–evaluate–improve loop. Quantitatively analyse errors, fix data pipeline issues, refine the model, and retrain until SER stabilises.

This sprint implements the **iterative refinement cycle** at the core of the agile methodology: after each training run, errors are analysed, root causes identified (data corruption, architecture limitations, filtering gaps), and targeted fixes applied before retraining. This cycle may execute multiple times within the sprint.

T4.1 - Error analysis of initial run (10 h)
Detailed breakdown of errors by type (deletion, substitution, insertion) and by token. Identify systematic patterns: which symbols are most confused and which image characteristics correlate with high SER.
Dependencies: T3.5.

T4.2 - Data pipeline improvements (15 h)
Fix data quality issues discovered during error analysis: annotation errors, problematic sample filtering, and vocabulary adjustments. Re-generate or re-filter the dataset as needed.
Dependencies: T4.1.

T4.3 - Model refinement and retraining (25 h)
Adjust hyperparameters based on error analysis and retrain on the improved dataset. Execute several training runs, each informed by the previous run's error profile, until SER stabilises.
Dependencies: T4.2.
Resources: Standard + consumer GPU.

T4.4 - Ablation studies (15 h)
Conduct controlled experiments to isolate the effect of key design choices in the data pipeline, model architecture, and training configuration. Document results in a structured ablation table.
Dependencies: T4.3.
Resources: Standard + consumer GPU.

T4.5 - Sprint 4 review (3 h)
Present the refined model metrics and ablation results to the supervisor. Decide whether further iteration is needed or whether to proceed to final evaluation.
Dependencies: T4.4.

WP4 subtotal: 68 h

2.6 WP5 - Sprint 5: Final Evaluation, Extension, and Defence

Duration: Weeks 17–21 (May – June 2026).
Sprint goal: Produce the definitive evaluation, explore conditional extensions, complete the thesis document, and prepare the defence.

T5.1 - Comprehensive evaluation and benchmarking (15 h)
Evaluate the best model on the held-out test set. Compute aggregate and per-category SER. Compare quantitatively against the classical CV baseline from Sprint 1. Produce publication-quality tables and figures.
Dependencies: T4.5.
Resources: Standard + consumer GPU.

T5.2 - Qualitative error analysis and case studies (10 h)
Select representative correct and incorrect predictions for qualitative discussion. Analyse failure modes by image difficulty (dense passages, unusual symbols, heavily augmented images). Produce annotated visualisations.
Dependencies: T5.1.

T5.3 - Extension exploration - conditional (20 h)
If the monophonic model achieves satisfactory SER (<10%) and time permits: prototype

a limited extension, such as chord symbol recognition or page-level staff detection. This task is **scope-contingent** and may be reduced or skipped without affecting the core deliverables.
Dependencies: T5.1.
Resources: Standard + consumer GPU.

T5.4 - Final thesis writing and revision (30 h)
Complete all thesis chapters: introduction, state of the art, methodology, dataset description, model architecture, experiments and results, conclusions, and future work. Integrate all figures, tables, and references. Perform full proofreading and formatting pass.
Dependencies: T5.2 (and T5.3 if completed).
Resources: Standard + LaTeX (TeX Live), Zotero.

T5.5 - Defence preparation (10 h)
Prepare the presentation slides, rehearse the defence talk, and prepare answers for anticipated questions from the evaluation committee.
Dependencies: T5.4.

WP5 subtotal: 85 h

2.7 Effort Summary

The total estimated effort is **509 hours**, distributed as shown in Table 2.

3 Gantt Chart

The Gantt chart in Figure 1 provides a visual representation of the project schedule. Key features:

- **Concurrency:** WP0 tasks (documentation, meetings) run throughout the project, avoiding a fully sequential plan.
- **Sprint structure:** Each sprint is clearly delimited, with a review milestone at the end.
- **Iterative loop:** Sprints 3 and 4 form the train–evaluate–improve cycle; Sprint 4 may revisit Sprint 3 activities.
- **Milestones** (red diamonds): Sprint reviews and key deliverables.

Table 2: Task summary with estimated effort, dependencies, and resources.

Code	Task	Hours	Dependencies	Specific Resources
WP0 - Project Management (130 h)				
T0.1	Context and scope definition	20	-	Zotero
T0.2	Temporal planning	12	T0.1	-
T0.3	Budget and sustainability	15	T0.2	-
T0.4	Final GEP integration	8	T0.3	-
T0.5	Supervisor meetings	25	Concurrent	-
T0.6	Thesis writing	50	Concurrent	LaTeX
WP1 - Sprint 1: Literature Review & Baseline (78 h)				
T1.1	OMR literature survey	25	-	Library access
T1.2	Dataset exploration	15	T1.1†	-
T1.3	Classical CV baseline impl.	25	T1.2	OpenCV
T1.4	Baseline evaluation and report	10	T1.3	-
T1.5	Sprint 1 review	3	T1.4	-
WP2 - Sprint 2: Dataset Construction (73 h)				
T2.1	Output encoding design	12	T1.5	music21
T2.2	Synthetic rendering pipeline	35	T2.1	LilyPond
T2.3	Data augmentation pipeline	15	T2.2	OpenCV
T2.4	Dataset validation and splits	8	T2.3	-
T2.5	Sprint 2 review	3	T2.4	-
WP3 - Sprint 3: CRNN-CTC Development (75 h)				
T3.1	CRNN architecture design	25	T2.5	Consumer GPU
T3.2	Training pipeline impl.	20	T3.1	-
T3.3	Initial training run	15	T3.2	GPU
T3.4	Evaluation framework	12	T3.2†	-
T3.5	Sprint 3 review	3	T3.3, T3.4	-
WP4 - Sprint 4: Iterative Refinement (68 h)				
T4.1	Error analysis of initial run	10	T3.5	-
T4.2	Data pipeline improvements	15	T4.1	-
T4.3	Model refinement runs	25	T4.2	GPU
T4.4	Ablation studies	15	T4.3	GPU
T4.5	Sprint 4 review	3	T4.4	-
WP5 - Sprint 5: Final Evaluation & Defence (85 h)				
T5.1	Comprehensive evaluation	15	T4.5	GPU
T5.2	Qualitative error analysis	10	T5.1	-
T5.3	Extension exploration (cond.)	20	T5.1	GPU
T5.4	Final thesis writing	30	T5.2	LaTeX
T5.5	Defence preparation	10	T5.4	-
	TOTAL	**509**		

†Partial overlap allowed; dependency is soft (task may start before predecessor finishes).

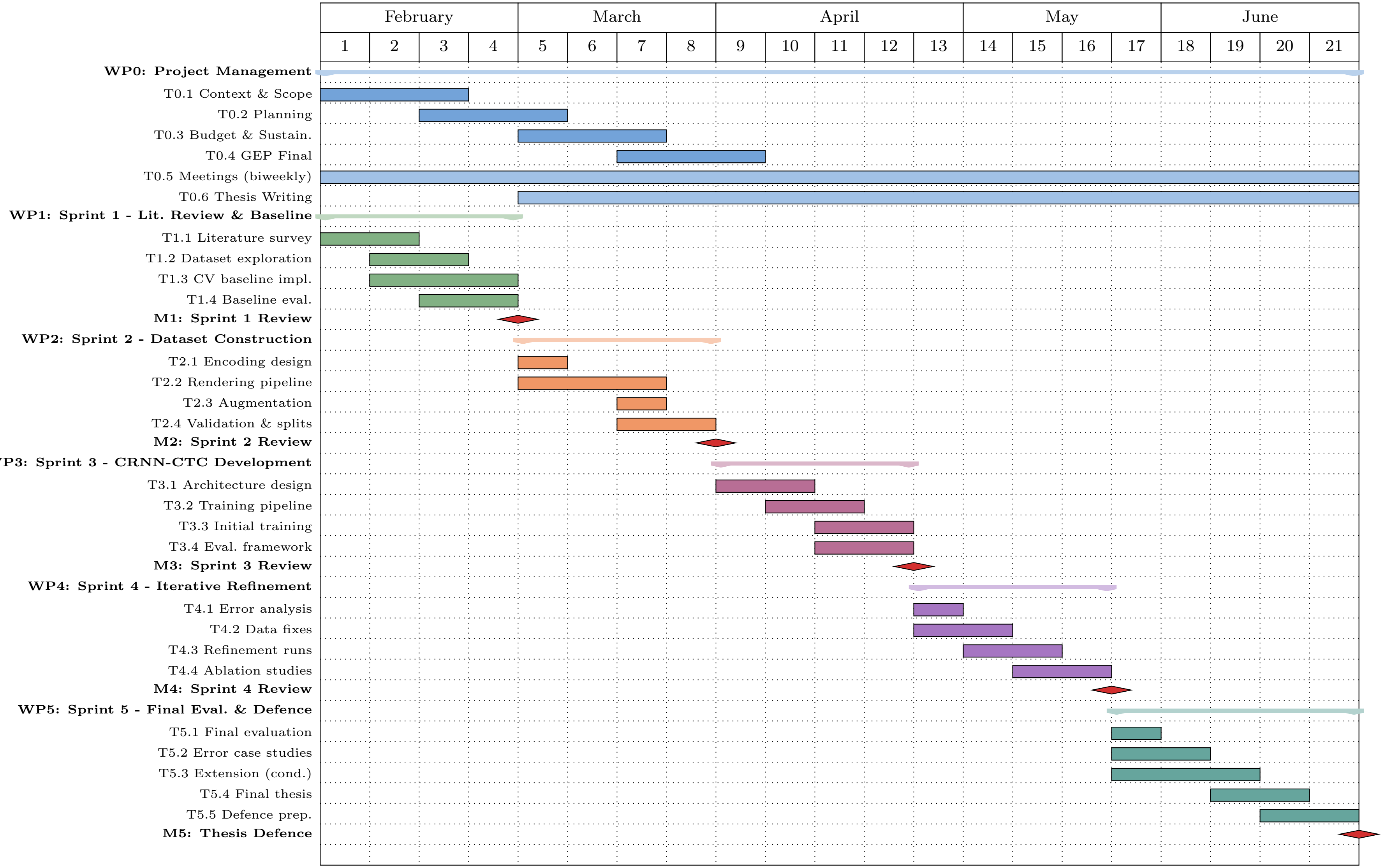

Figure 1: Project Gantt chart (weeks 1–21, February–June 2026). Coloured bars indicate work packages; red diamonds mark sprint reviews and the final defence milestone. WP0 (project management, meetings, and thesis writing) runs concurrently throughout.

4 Risk Management and Contingency Plans

This section coordinates with the risks identified in Deliverable 1 (Section 3.6) and defines concrete contingency actions, their impact on the schedule, and the additional resources required.

4.1 R1 - Limited Ground-Truth Data

Risk: Annotated Real Book data is not publicly available. The synthetic dataset may contain errors or lack diversity.
Likelihood: High. **Impact:** High (degraded model accuracy).
Mitigation (built into plan): T2.2 and T2.3 dedicate 50 hours to synthetic generation and augmentation, including iterative quality checks (T2.4). An overestimation of 5 hours is embedded in T2.2 to handle rendering edge cases.
Contingency (Plan B): If the LilyPond pipeline produces unusable output for >10% of samples, fall back to training exclusively on the original PrIMuS typeset images without jazz-style rendering. This eliminates T2.2 entirely but sacrifices visual domain similarity to the Real Book.

- *Schedule impact:* Saves ∼35 h from T2.2 if activated early; may require ∼10 h of additional augmentation effort in T2.3 to bridge the domain gap.
- *Resources:* No additional resources; PrIMuS data is already downloaded.

4.2 R2 - Staff Detection Errors Propagating Downstream

Risk: Incorrectly cropped staff lines feed corrupted input to the CRNN, increasing SER.
Likelihood: Medium. **Impact:** Medium.
Mitigation (built into plan): The current pipeline assumes pre-cropped lines (synthetic data is generated at line level). This risk is relevant only for the extension (T5.3) targeting full Real Book pages.
Contingency (Plan B): If page-level detection is attempted in T5.3 and fails, restrict the demonstration to manually cropped regions. This does not affect the core deliverables.

- *Schedule impact:* T5.3 hours reassigned to thesis writing (T5.4).
- *Resources:* None additional.

4.3 R3 - Model Underfitting or CTC Misalignment

Risk: The CRNN does not converge, or CTC loss plateaus at an unacceptably high SER (>30%).
Likelihood: Medium. **Impact:** High (could compromise the project's central contribution).
Mitigation (built into plan): Sprint 4 (68 h) is entirely dedicated to iterative refinement. T4.1–T4.3 form a systematic analyse–fix–retrain loop. An overestimation of ∼5 h is embedded in T4.3 to allow additional runs.
Contingency (Plan B): If after Sprint 4 the model still exceeds 20% SER, simplify the task: reduce vocabulary to pitch-only (removing durations), train on clean (non-augmented) images, or increase the image height to provide more visual detail. These simplifications can be implemented within T4.2–T4.3 without new tasks.

- *Schedule impact:* Up to 10 h from T5.3 (extension) can be reallocated to additional refinement iterations.
- *Resources:* Same hardware; longer GPU time per run.

4.4 R4 - Computational Resource Constraints

Risk: Training runs take too long on the available hardware, slowing the iteration cycle and delaying Sprint 4.
Likelihood: Low–Medium. **Impact:** Medium (schedule delays).
Mitigation (built into plan): The model is designed to be lightweight and uses mixed-precision

training to reduce memory usage and training time.
Contingency (Plan B): If training time becomes a bottleneck, reduce dataset size, simplify the model architecture, or request temporary access to UPC or cloud GPU resources.

- *Schedule impact:* Minimal if mitigated early; up to 5 h overhead for cloud setup.
- *Resources:* Potential cloud GPU cost (∼€20–50 for short-term rental).

4.5 R5 - Scope Creep Toward Polyphony

Risk: Attempting polyphonic extensions prematurely diverts effort from the monophonic core.
Likelihood: Medium. **Impact:** High (incomplete core system).
Mitigation (built into plan): Polyphonic extension is isolated in T5.3, explicitly labelled as "conditional," and scheduled *after* final evaluation (T5.1). It cannot start until the monophonic model achieves <10% SER.
Contingency (Plan B): If time runs short, T5.3 is reduced to a brief feasibility discussion in the thesis (2–3 hours of writing), and the remaining hours are redirected to T5.4 (thesis completion).

- *Schedule impact:* Up to 18 h freed from T5.3 and added to T5.4.
- *Resources:* None additional.

4.6 Summary of Schedule Buffers

The plan includes the following built-in margins to absorb deviations:

- **Overestimation in critical tasks:** T2.2 (+5 h), T4.3 (+5 h).
- **Conditional extension:** T5.3 (20 h) serves as a flexible buffer that can be fully reallocated to refinement or writing.
- **Thesis writing throughout:** T0.6 runs concurrently, preventing a documentation bottleneck at the end.
- **Total slack:** Approximately 25–30 h can be redistributed without exceeding the project end date.

References

[1] C. Dalitz, M. Droettboom, B. Pranzas, and I. Fujinaga, "A Comparative Study of Staff Removal Algorithms," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 30, no. 5, pp. 753–766, May 2008, ISSN: 0162-8828. DOI: 10.1109/TPAMI.2007.70749. Accessed: Feb. 18, 2026. [Online]. Available: http://ieeexplore.ieee.org/document/4359356/.

[2] "PrIMuS dataset," Accessed: Feb. 18, 2026. [Online]. Available: https://grfia.dlsi.ua.es/primus/.

[3] A. R. Vila, *Antoniorv6/SMT*, Feb. 14, 2026. Accessed: Feb. 19, 2026. [Online]. Available: https://github.com/antoniorv6/SMT.

[4] "Ufal/olimpic-icdar24: Practical End-to-End Optical Music Recognition for Pianoform Music," Accessed: Feb. 18, 2026. [Online]. Available: https://github.com/ufal/olimpic-icdar24?tab=readme-ov-file.